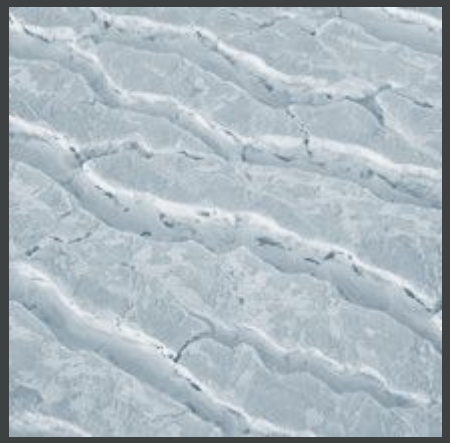

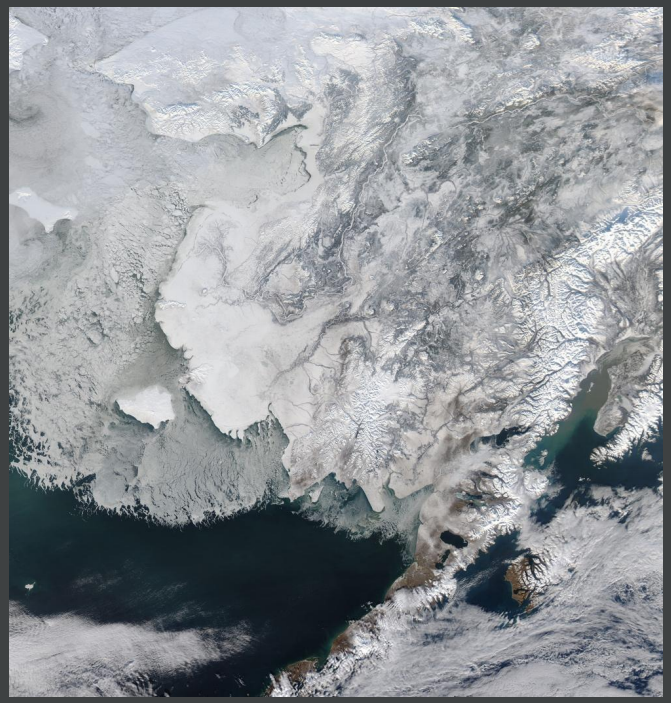

A Retrospective Application of Research Metadata Schema WG Recommendations

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NASA National Snow and Ice Data Center Distributed Active Archive Center

RDA VP17 Research Metadata Schemas WG
22 April 2021

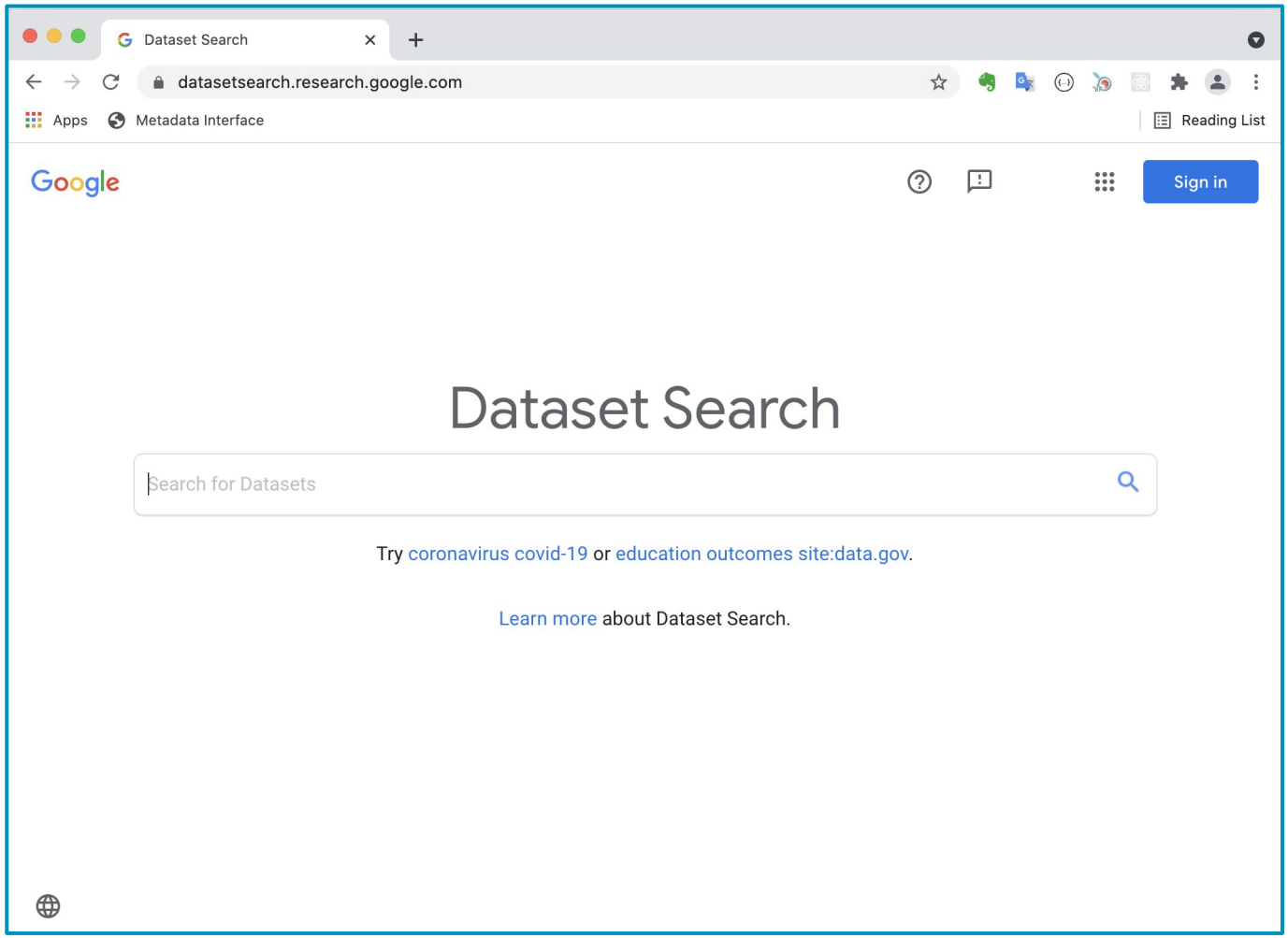
Background

- NSIDC manages "landing page" content for our research data catalog using Drupal CMS (currently Drupal v7).
- We added schema.org markup to the Drupal CMS workflow for our dataset landing pages in late 2019.
- How many of the recommendations did we follow?
- How would addressing the missed recommendations improve our results?

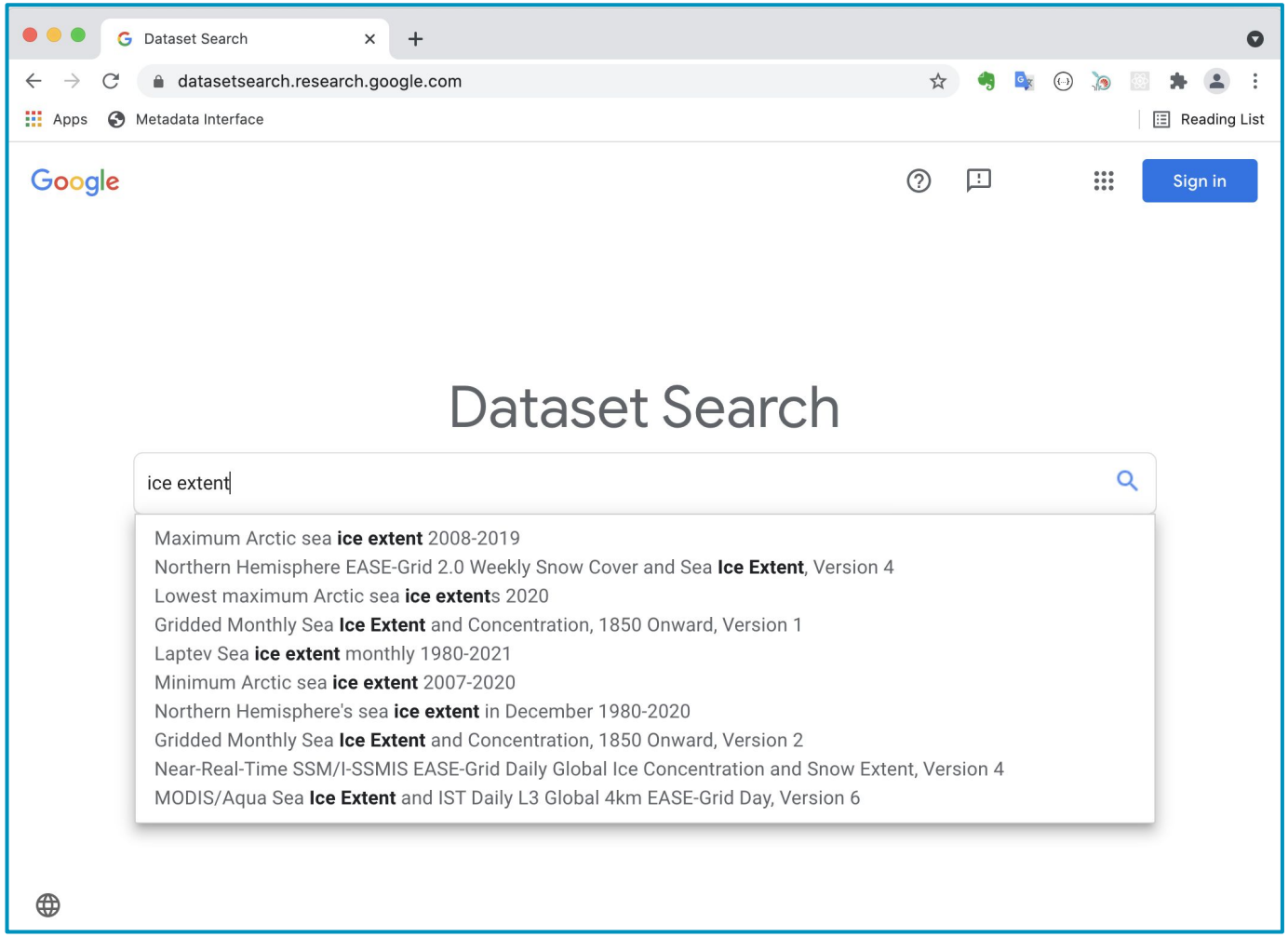
#1: Clarify the purpose(s)

Increase Search Engine Optimization for NSIDC dataset landing pages, specifically with regard to Google Dataset Search.

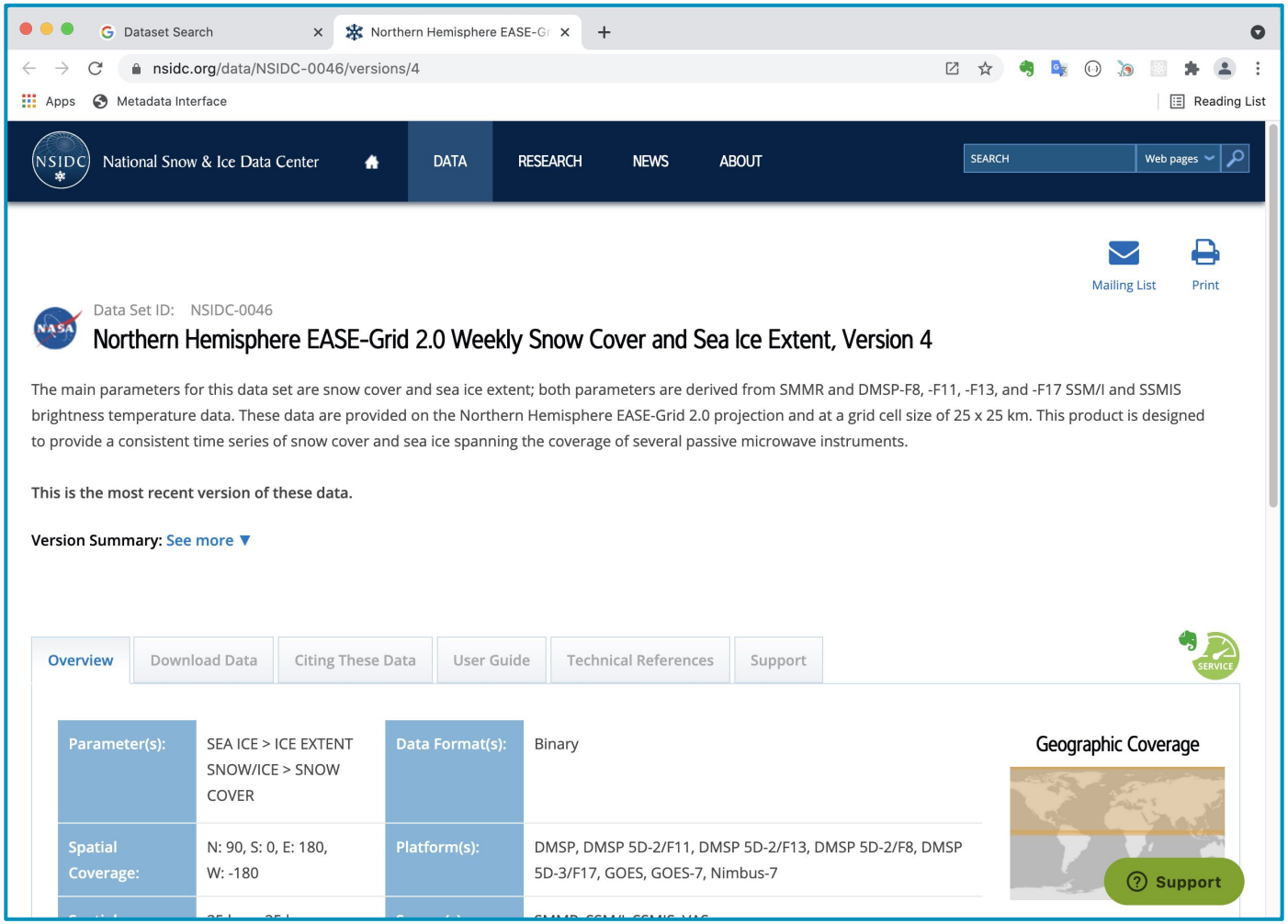
#1: Clarify the purpose(s)



#1: Clarify the purpose(s)



#2: Identify resource(s) to be added



#3: Adopt/develop a crosswalk

Generated simple mapping:

DIF fields	Drupal Dataset Field Tokens	schema.org
Entry_Title	[node:title] | [site:name]	Page title
Abstract	[node:body]	Dataset description
Keyword	[node:field_dataset_parameter_gcmd]	Keywords
Dataset_Series_Name	[node:field_dataset_id]	ID
Entry_Title	[node:title]	Name
Data_Set_Citation/Version	[node:field_dataset_version]	Version

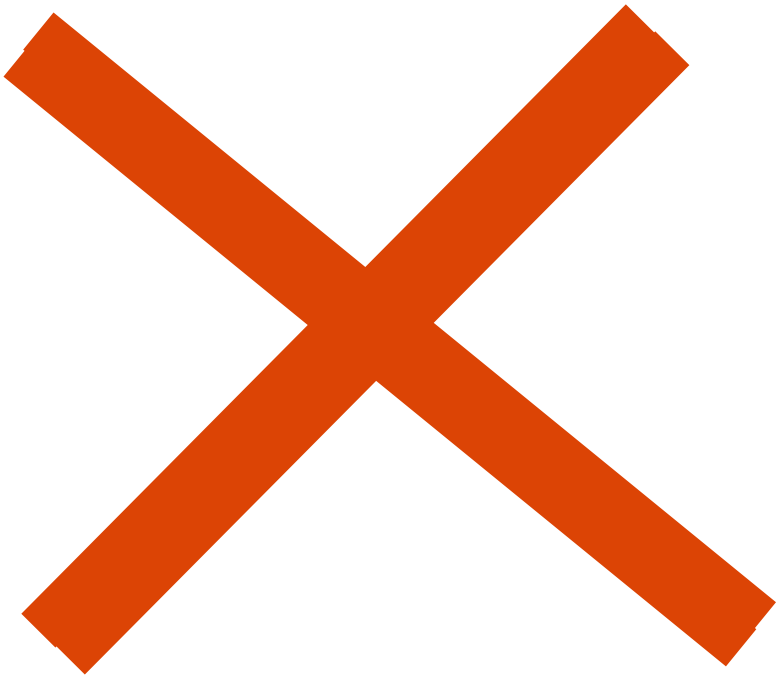
#3: Adopt/develop a crosswalk

DIF fields	Drupal Dataset Field Tokens	schema.org
Temporal_Coverage /Start_Date Temporal_Coverage /End_Date	[node:field_dataset_temporal_coverage]	Temporal Coverage
Spatial_Coverage/Southernmost_Latitude Spatial_Coverage/Easternmost_Latitude Spatial_Coverage/Westernmost_Latitude Spatial_Coverage/Northernmost_Latitude	[node:field_dataset_geo_coordinates]	Spatial Coverage
Data_Set_Citation/Dataset_Creator	[node:field_dataset_citation_link]	Identifier (DOI)
	https://nsidc.org/data/[node:field_dataset_id]/versions/[node:field_dataset_version]	URL
	https://nsidc.org	Organization id

#3: Adopt/develop a crosswalk

DIF fields	Drupal Dataset Field Tokens	schema.org
PROVIDER (the provider of the dataset)		
Data_Set_Citation/Dataset_Publisher	[node:field_dataset_citation_publisher]	Name
	https://nsidc.org/data/[node:field_dataset_id]/versions/[node:field_dataset_version]	URL (absolute URL of the canonical Web page)
AUTHOR (the author(s) of the dataset)		
Personnel/Role/Investigator/First_Name Personnel/Role/Investigator/Last_Name	[node:field_dataset_investigator_name]	Name
PUBLISHER (the publisher of the dataset)		
	National Snow and Ice Data Center	Name
	https://nsidc.org	Organization id

#4: Incorporate external vocabulary



#5: Implement markup syntax consistently

- We were already building landing page content from dynamically-generated DIF metadata records.
- The content of the DIF records is parsed and stored locally by Drupal.
- The Drupal **`Schema.org Metatag`** module is used to map the DIF content to JSON-LD in the landing page HTML output.

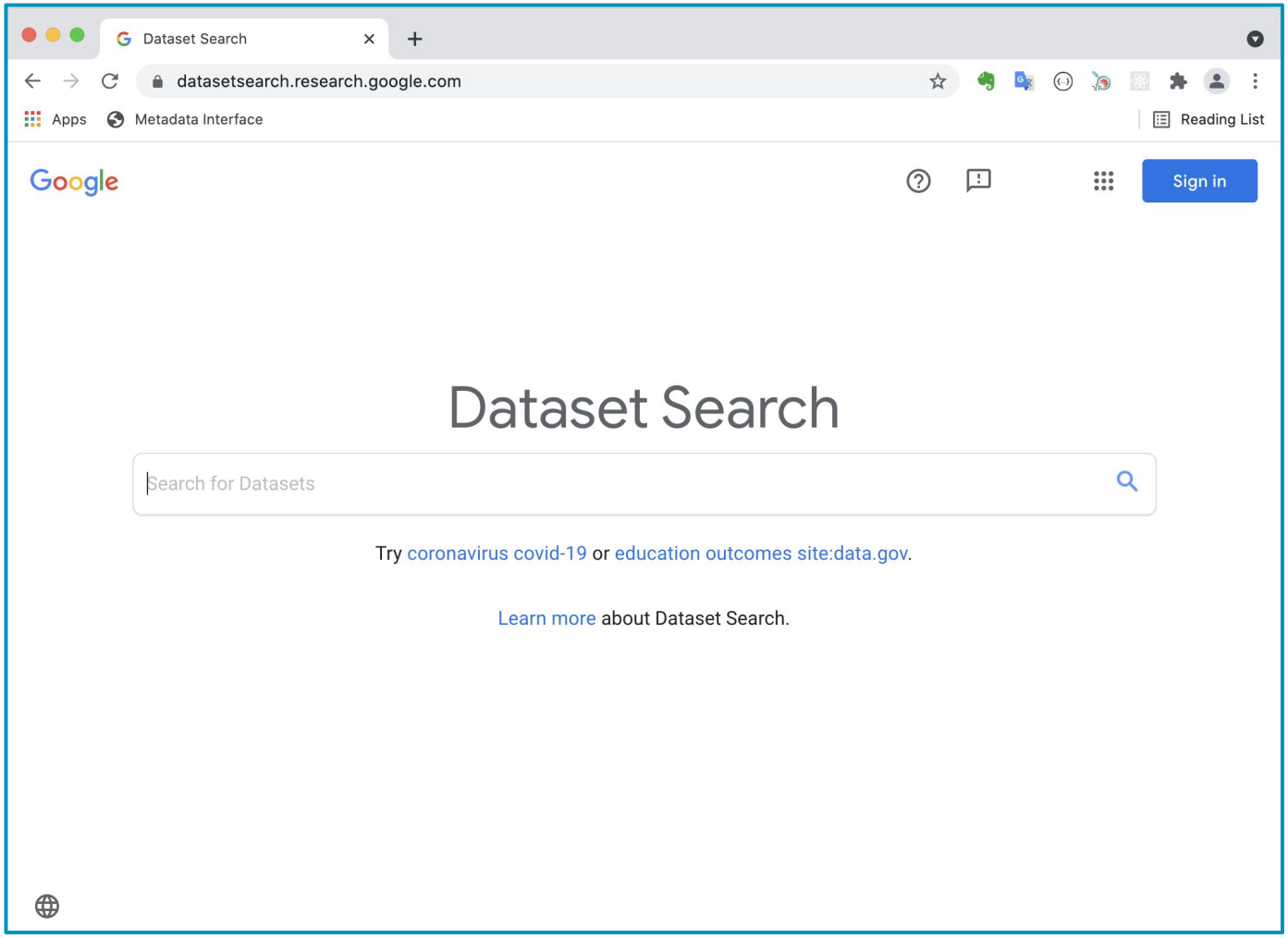
#6: Be friendly to web crawlers

No sitemap yet; will be implemented with transition to Drupal 8 in late summer 2021.

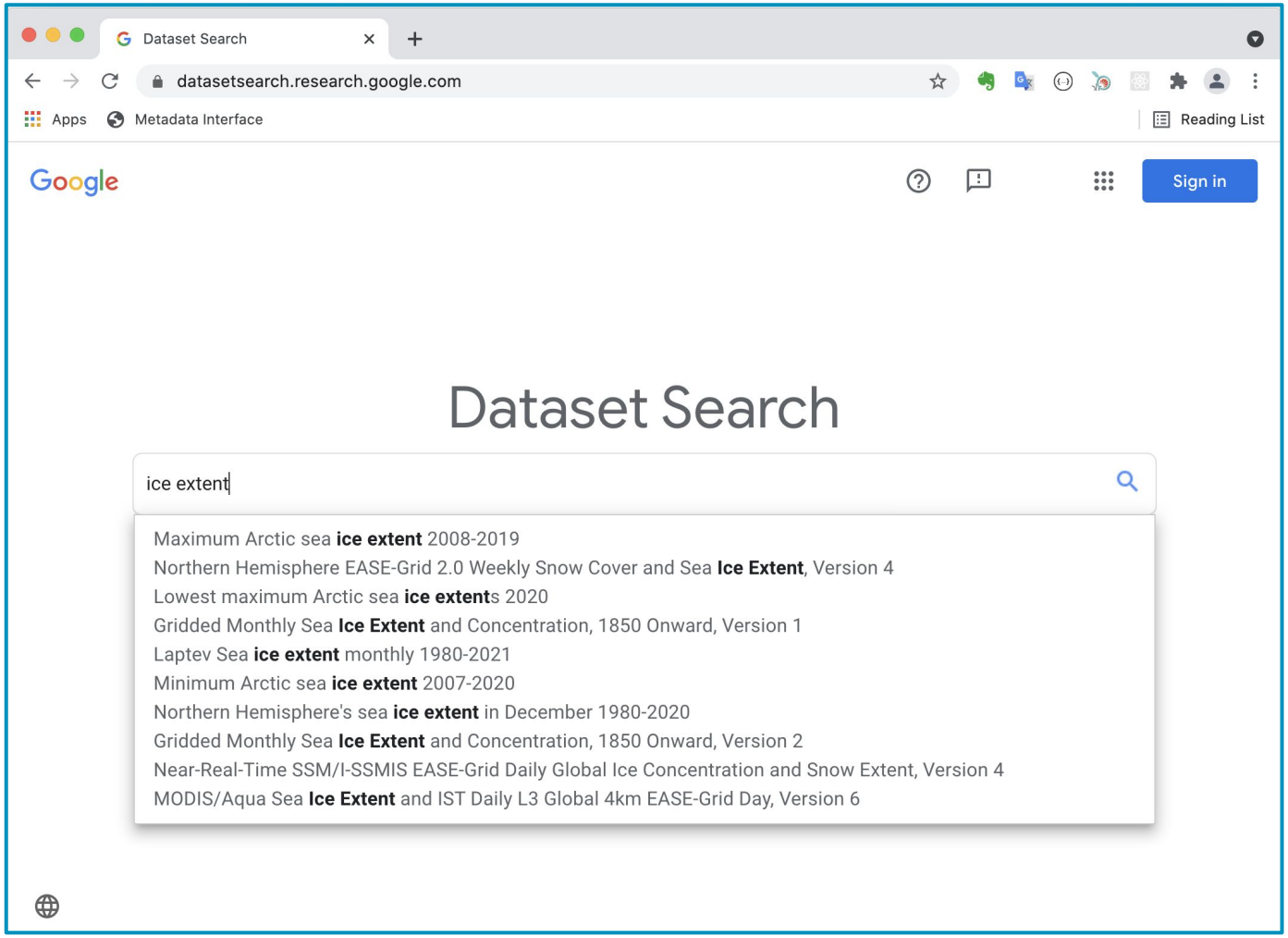
#7: Make the best use of available tools

- Database-driven metadata record generation and Drupal CMS help with consistency.
- *Need to explore validation tools, others.*

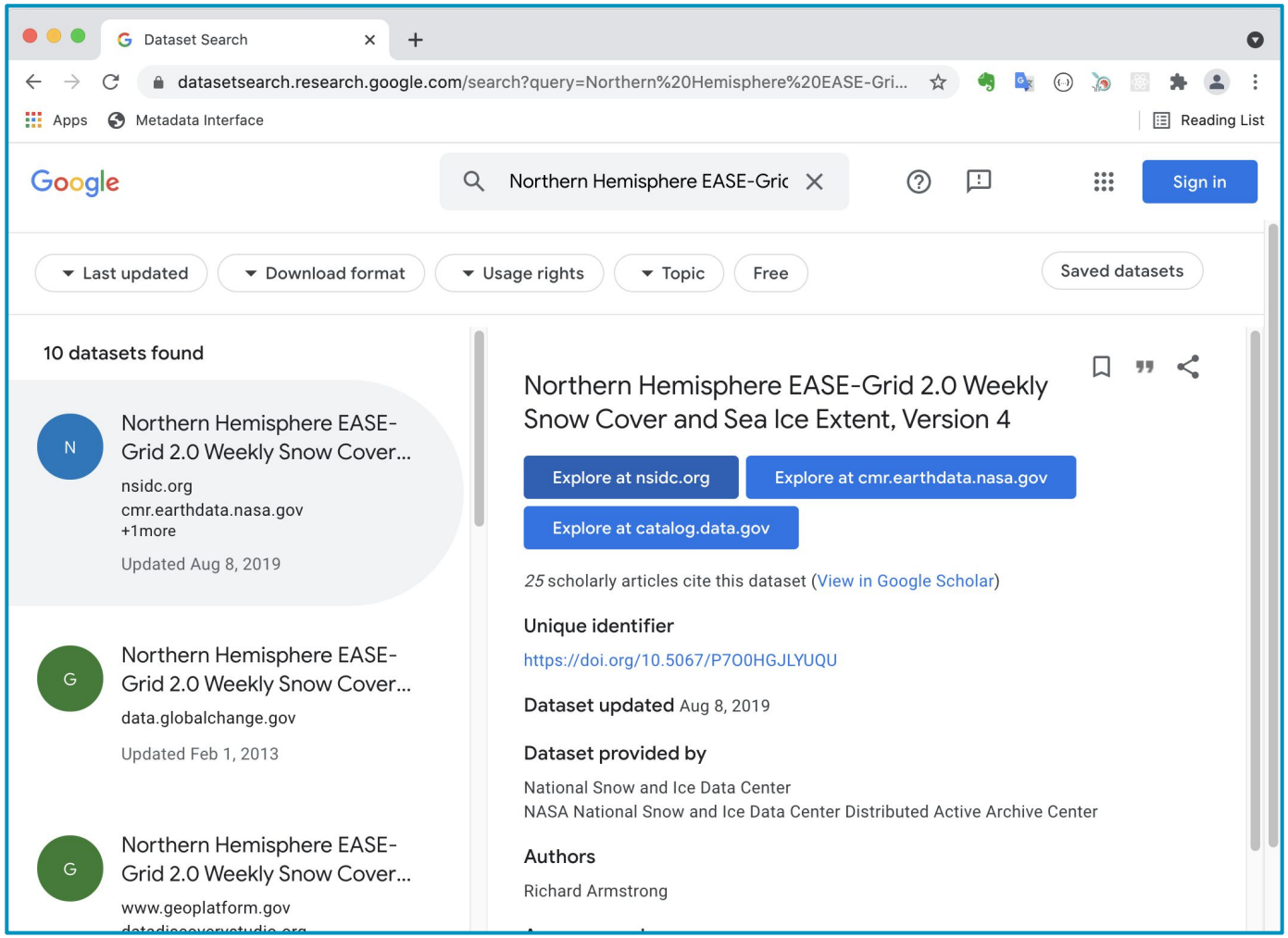
Results


Dataset Search
datasetsearch.research.google.com
Apps
Metadata Interface
Reading List
Google
Sign in
Dataset Search
Search for Datasets
Try coronavirus covid-19 or education outcomes site:data.gov.
Learn more about Dataset Search.

Results


Dataset Search
datasetsearch.research.google.com
Apps
Metadata Interface
Reading List
Google
Sign in
Dataset Search
ice extent
Maximum Arctic sea ice extent 2008-2019
Northern Hemisphere EASE-Grid 2.0 Weekly Snow Cover and Sea Ice Extent, Version 4
Lowest maximum Arctic sea ice extents 2020
Gridded Monthly Sea Ice Extent and Concentration, 1850 Onward, Version 1
Laptev Sea ice extent monthly 1980-2021
Minimum Arctic sea ice extent 2007-2020
Northern Hemisphere's sea ice extent in December 1980-2020
Gridded Monthly Sea Ice Extent and Concentration, 1850 Onward, Version 2
Near-Real-Time SSM/I-SSMIS EASE-Grid Daily Global Ice Concentration and Snow Extent, Version 4
MODIS/Aqua Sea Ice Extent and IST Daily L3 Global 4km EASE-Grid Day, Version 6

Results


Dataset Search
datasetsearch.research.google.com/search?query=Northern%20Hemisphere%20EASE-Gri...
Apps
Metadata Interface
Reading List
Google
Northern Hemisphere EASE-Gric
Sign in
Last updated
Download format
Usage rights
Topic
Free
Saved datasets
10 datasets found
N
Northern Hemisphere EASE-Grid 2.0 Weekly Snow Cover...
nsidc.org
cmr.earthdata.nasa.gov
+1more
Updated Aug 8, 2019
G
Northern Hemisphere EASE-Grid 2.0 Weekly Snow Cover...
data.globalchange.gov
Updated Feb 1, 2013
G
Northern Hemisphere EASE-Grid 2.0 Weekly Snow Cover...
www.geoplatform.gov
Northern Hemisphere EASE-Grid 2.0 Weekly Snow Cover and Sea Ice Extent, Version 4
Explore at nsidc.org
Explore at cmr.earthdata.nasa.gov
Explore at catalog.data.gov
25 scholarly articles cite this dataset (View in Google Scholar)
Unique identifier
https://doi.org/10.5067/P7O0HGJLYUQU
Dataset updated Aug 8, 2019
Dataset provided by
National Snow and Ice Data Center
NASA National Snow and Ice Data Center Distributed Active Archive Center
Authors
Richard Armstrong

Results

```
<script type="application/ld+json">{
    "@context": "http://schema.org",
    "@graph": [
        {
            "@type": "Dataset",
            "provider": {
                "@type": "Organization",
                "name": "NASA National Snow and Ice Data Center Distribut
                "url": "https://nsidc.org/data/NSIDC-0046/versions/4",
                "logo": {
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                    "representativeOfPage": "True",
                    "url": "https://nsidc.org/sites/nsidc.org/files/image
                    "width": "60 px",
                    "height": "60 px"
                }
            },
            "@id": "NSIDC-0046",
            "name": "Northern Hemisphere EASE-Grid 2.0 Weekly Snow Cover
            "version": "4",
```

#8: Document and share every step

Partial success: Some internal documentation, but not enough!

#9: Find and join a community

- Arctic Data Committee/POLDER (https://github.com/POLDER-Crew)
- *Improve connection to ESIP Science-on-Schema.org activities.*

Where we connected with the recommendations

- #1: Clarify the purpose
- #2: Identify resources to be added
- #3: Adopt/develop a crosswalk
- #5: Implement markup syntax consistently
- #7: Make the best use of available tools

Where we didn't connect…

- #4: Incorporate external vocabulary
- #6: Be friendly to web crawlers
- #8: Document and share every step
- #9: Find and join a community

Thanks to NSIDC colleagues:

- Shannon Leslie
- Amanda Leon
- Mary Ellen Byers

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